

Supporting Information for Extra Hidden States under Emission Misspecification in Hidden Markov Models

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S1 Proofs

This section gives the proofs of Lemma 2.3, Propositions 2.4–2.5, Theorem 2.6, and Corollary 2.7 from Section 2 of the main article, in that order. Notation is identical to that article. For convenience, the Kullback–Leibler divergence and $L^1(\lambda)$ distance used below are $D(p\|q) = \int p \log(p/q) d\lambda$ and $\|p - q\|_1 = \int |p - q| d\lambda$, with the extended-value conventions stated in the main article. Also let $\Delta_K = \{(\omega_1, \dots, \omega_K) \in [0, 1]^K : \sum_{j=1}^K \omega_j = 1\}$ denote the K -weight probability simplex.

S1.1 Proof of Lemma 2.3: separation from lower-order mixtures

The class $\mathcal{M}_{K_0}$ is the continuous image of the compact set $\Delta_{K_0} \times \Theta^{K_0}$ and is therefore compact in L^1 . The constrained simplex defining $\mathcal{A}_c$ is compact, so $\mathcal{A}_c$ is compact as well. The sets are disjoint. Otherwise, a mixture containing exactly R positive-weight, pairwise distinct atoms would equal a mixture with at most K_0 atoms after zero-weight and repeated atoms were removed. This contradicts identifiable minimal order. Two disjoint compact sets in a metric space have positive distance, hence

$$d_c = \inf_{p \in \mathcal{A}_c} \inf_{q \in \mathcal{M}_{K_0}} \|p - q\|_1 > 0.$$

Pinsker’s inequality, $D(p\|q) \geq \|p - q\|_1^2 / 2$, then gives the claimed Kullback–Leibler lower bound.

S1.2 Proof of Proposition 2.4: aligned improvement by state splitting

In the aligned comparison, the hidden aggregate trajectory law is held fixed. All transition terms therefore cancel in the complete-data likelihood ratio. With one fitted emission f_{η_k} per state, stationarity gives the risk

$$\sum_{k=1}^{K_0} \pi_k D(g_k \| f_{\eta_k}).$$

Taking the infimum over the emission parameters gives $\sum_k \pi_k \delta_k(1)$. With m_k substates, the fitted aggregate emission can be any $q_k \in \mathcal{M}_{m_k}$, yielding $\sum_k \pi_k \delta_k(m_k)$. Since $\mathcal{M}_1 \subseteq \mathcal{M}_{m_k}$, all terms

weakly decrease. A strict decrease in one state with $\pi_k > 0$ proves the claim.

S1.3 Proof of Proposition 2.5: exact HMM refinement

Let X_t^{ref} denote the refined chain and let $r(k, a) = k$ map a refined state to its aggregate state. The refined transition matrix is stochastic because

$$\sum_{\ell, b} \gamma_{k\ell}^0 \omega_{\ell, b} = 1.$$

The distribution $\pi_k \omega_{k, a}$ is stationary since

$$\sum_{k, a} \pi_k \omega_{k, a} \gamma_{k\ell}^0 \omega_{\ell, b} = \omega_{\ell, b} \sum_k \pi_k \gamma_{k\ell}^0 = \pi_{\ell} \omega_{\ell, b}.$$

Furthermore,

$$\mathbb{P}\{r(X_t^{\text{ref}}) = \ell \mid X_{t-1}^{\text{ref}} = (k, a)\} = \sum_b \gamma_{k\ell}^0 \omega_{\ell, b} = \gamma_{k\ell}^0.$$

Summing the refined joint density over substate paths compatible with a fixed aggregate path $z_{1:T} = (z_1, \dots, z_T)$ gives

$$\begin{aligned} \pi_{z_1} \prod_{t=2}^T \gamma_{z_{t-1}, z_t}^0 \prod_{t=1}^T \sum_{a=1}^{m_{z_t}} \omega_{z_t, a} f_{\eta_{z_t, a}^0}(y_t) \\ = \pi_{z_1} \prod_{t=2}^T \gamma_{z_{t-1}, z_t}^0 \prod_{t=1}^T g_{z_t}(y_t). \end{aligned}$$

Summing over aggregate paths proves equality of all observed finite-dimensional distributions.

S1.4 Proof of Theorem 2.6: observed-law separation

Exact refinement establishes

$$\inf_{Q \in \mathbf{H}_R(\mathcal{F})} \underline{D}^{\text{rate}}(P_0^Y \| Q) = 0.$$

Write $\mathbb{P}_0$ and $\mathbb{E}_0$ for probability and expectation under the true aggregate HMM. Fix a candidate $Q \in \mathbf{H}_{K_0}(\mathcal{F})$ with emissions $f_{\eta_1}, \dots, f_{\eta_{K_0}}$. Let $\beta_{t,j}^Q(y_{1:t-1})$ be its predictive probability of hidden state j given the realized observation history $y_{1:t-1} = (y_1, \dots, y_{t-1})$. Its one-step predictive density has the form

$$p_t^Q(y \mid y_{1:t-1}) = \sum_{j=1}^{K_0} \beta_{t,j}^Q(y_{1:t-1}) f_{\eta_j}(y),$$

where the $\beta_{t,j}^Q$ are nonnegative and sum to one. Thus the predictive density belongs to $\mathcal{M}_{K_0}$ for every history, regardless of the candidate transition matrix and initial distribution.

For the true aggregate HMM, let

$$\beta_{t,k}^0(y_{1:t-1}) = \mathbb{P}_0(Z_t = k \mid Y_{1:t-1} = y_{1:t-1}).$$

Stationarity implies $\pi_\ell = \sum_k \pi_k \gamma_{k\ell}^0 \geq \varepsilon$, so $\beta_{1,\ell}^0 \geq \varepsilon$. For $t \geq 2$,

$$\begin{aligned} \beta_{t,\ell}^0(y_{1:t-1}) &= \sum_k \mathbb{P}_0(Z_{t-1} = k \mid Y_{1:t-1} = y_{1:t-1}) \gamma_{k\ell}^0 \\ &\geq \varepsilon. \end{aligned}$$

The true predictive density is consequently

$$p_t^0(y \mid Y_{1:t-1}) = \sum_{k,a} \beta_{t,k}^0(Y_{1:t-1}) \omega_{k,a} f_{\eta_{k,a}^0}(y),$$

with every refined weight at least $c = \varepsilon \omega_{\min}$. Thus $p_t^0(\cdot \mid Y_{1:t-1}) \in \mathcal{A}_c$ almost surely. By the mixture-separation lemma,

$$D\{p_t^0(\cdot \mid Y_{1:t-1}) \parallel p_t^Q(\cdot \mid Y_{1:t-1})\} \geq d_c^2/2$$

almost surely. If a conditional divergence is infinite, the desired bound is immediate. Otherwise, the chain rule gives

$$\begin{aligned} D\{(P_0^Y)^{1:T} \parallel Q^{1:T}\} &= \sum_{t=1}^T \mathbb{E}_0 \left[D\{p_t^0(\cdot \mid Y_{1:t-1}) \parallel p_t^Q(\cdot \mid Y_{1:t-1})\} \right] \\ &\geq T d_c^2/2. \end{aligned}$$

Dividing by T and taking a lower limit proves the positive relative-entropy-rate bound.

S1.5 Proof of Corollary 2.7: enriched-emission representation

Each true emission g_k belongs to $\mathcal{M}_M$ when $M \geq \max_{1 \leq k \leq K_0} m_k$. The original aggregate chain is therefore an element of $\mathbf{H}_{K_0}(\mathcal{M}_M)$ with observed law P_0^Y , and the infimum of the relative-entropy rate is zero.

S2 Simulation computation and diagnostics

The reproducibility package accompanying this submission, named `4416781_Reproducibility_Package.zip`, contains the code, fixed seeds, data provenance, and generated outputs. From its extracted root, the fixed-seed simulation is run entirely in R by `numerics/scripts/run_review_simulation.R`, with master seed 20260610.

The replication script compares a Gaussian-emission HMM with $K = 2$, a Gaussian-emission HMM with $K = 3$, and a two-state HMM with two Gaussian components per state-dependent

distribution. The sample sizes are $T \in \{500, 1500, 5000\}$, with 200 replications per sample size and 10 random EM starts per replicated fit. All starts are screened deterministically and the three best are refined with relative log-likelihood tolerance 10^{-6} and at most 100 refinement iterations. The Gaussian-emission fits use 25 screening iterations and the mixture-emission fits use 30. The implementation uses scaled forward–backward recursions, a minimum standard deviation of 0.05, and deterministic ordering by state-level mean.

Model comparisons use the Bayesian information criterion (BIC), as defined in the main article. Row-level results are stored in `numerics/output/review_replications.csv`; the compact summary is in `numerics/output/review_summary_by_T.csv`. Additional diagnostics appear in Figures [S1–S3](#).

S3 Elk application details

The public elk locations are read directly from the `elk_data` object distributed with `moveHMM`. No copy of the data is redistributed in the reproducibility package. At run time, `numerics/scripts/run_elk_application.R` loads the object directly into memory and verifies its structure and SHA-256 digest before fitting any model. No raw-data CSV is created or redistributed. The archived results used `moveHMM` version 1.10. The file `empirical/DATA_SOURCE.md` records the data provenance, source publications, loading command, package licence, and expected object digest. The exact timestamps are unavailable, so no claims are made about variation in sampling intervals.

The script `numerics/scripts/run_elk_application.R` constructs consecutive Euclidean step lengths within each animal, converts coordinate distances from metres to kilometres, applies $\log(1+x)$ to each step length x , and fits the three models reported in the main article. Its master seed is 20260826. Each model uses 200 starts, 70 screening iterations, refinement of the 10 best starts, at most 500 refinement iterations, relative log-likelihood tolerance 10^{-6} , and a minimum standard deviation of 0.05. Output CSV files contain model comparisons, state and mixture parameters, transition matrices, Viterbi decodings, and decoded-state cross-tabulations; these are derived results, not redistributed raw locations. All figures are generated with `ggplot2`.

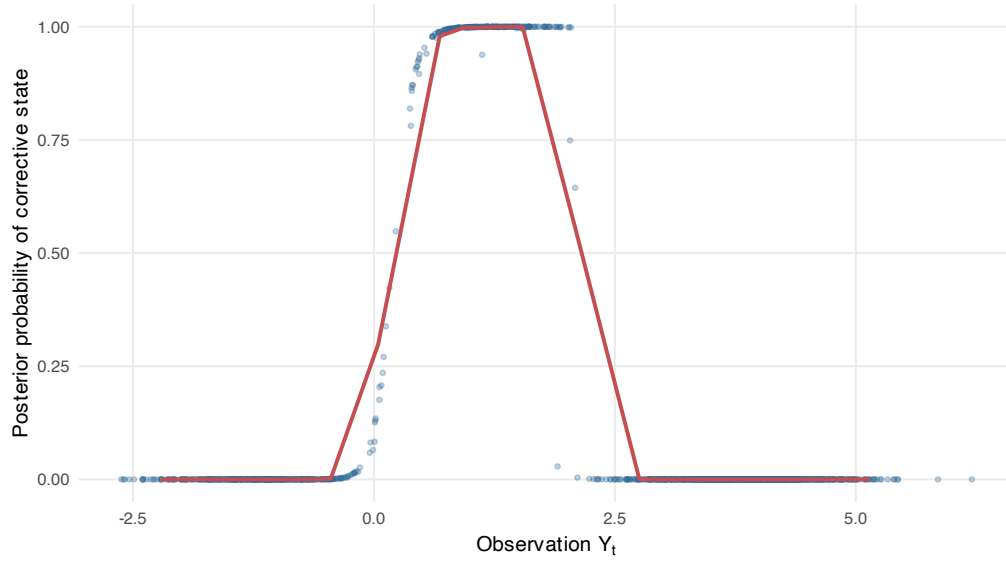


Figure S1: Posterior probability of the corrective Gaussian-HMM state in the reference $K = 3$ fit. The probability tracks observations associated with the second component of the heterogeneous first aggregate regime.

BIC favours $K=3$ in 94% of replications

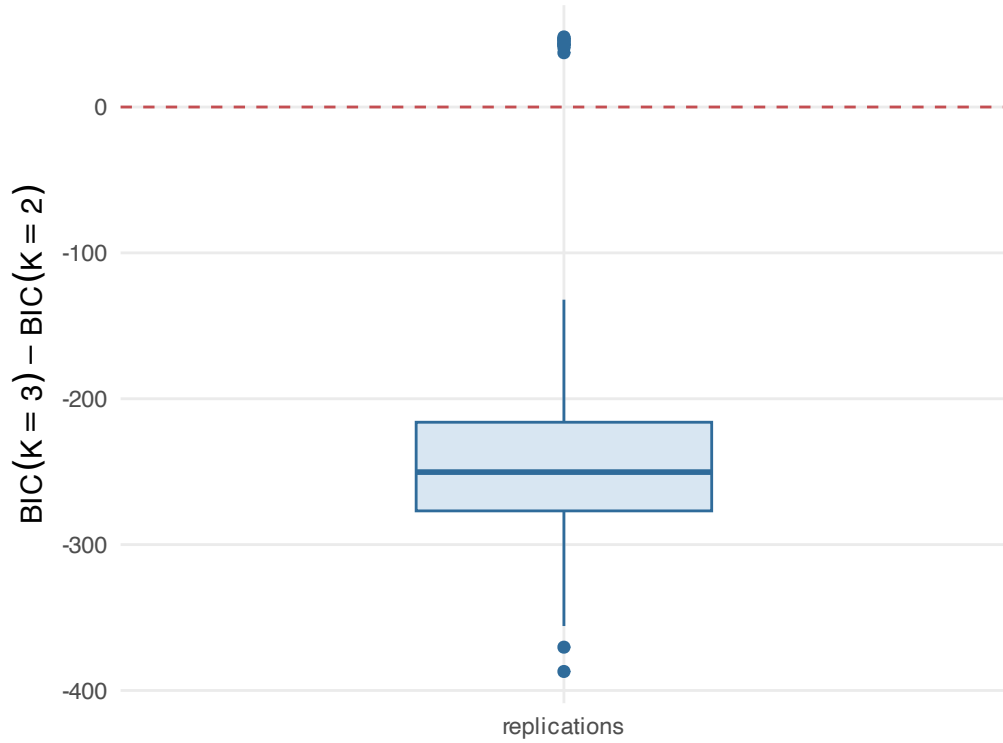


Figure S2: Distribution of the BIC difference between the three-state and two-state Gaussian models over the $T = 1500$ replications. Negative values favour the corrective three-state Gaussian HMM.

Table S1: One-step-ahead predictive calibration for the elk models. Smaller Kolmogorov–Smirnov (KS) distance from uniformity is better; nominal interval coverage is 0.90.

Animal	model	PIT KS distance	90% coverage
elk-115	Gaussian K=2	0.062	0.953
elk-115	Gaussian K=3	0.062	0.948
elk-115	Mixture K=2	0.043	0.922
elk-163	Gaussian K=2	0.087	0.924
elk-163	Gaussian K=3	0.044	0.930
elk-163	Mixture K=2	0.046	0.924
elk-287	Gaussian K=2	0.102	0.951
elk-287	Gaussian K=3	0.067	0.914
elk-287	Mixture K=2	0.048	0.914
elk-363	Gaussian K=2	0.064	0.926
elk-363	Gaussian K=3	0.050	0.940
elk-363	Mixture K=2	0.040	0.931

Table S2: State-dependent parameters for the enriched two-state elk models. Component medians are back-transformed to kilometres.

Animal	state	component	weight	median step	occupation	self-transition
elk-115	aggregate 1	1	0.20	0.06 km	0.84	0.90
elk-115	aggregate 1	2	0.80	0.59 km	0.84	0.90
elk-115	aggregate 2	1	0.70	2.60 km	0.16	0.46
elk-115	aggregate 2	2	0.30	9.38 km	0.16	0.46
elk-163	aggregate 1	1	0.77	0.07 km	0.62	0.76
elk-163	aggregate 1	2	0.23	0.27 km	0.62	0.76
elk-163	aggregate 2	1	0.45	0.58 km	0.38	0.60
elk-163	aggregate 2	2	0.55	5.26 km	0.38	0.60
elk-287	aggregate 1	1	0.57	0.18 km	0.44	1.00
elk-287	aggregate 1	2	0.43	0.60 km	0.44	1.00
elk-287	aggregate 2	1	0.58	0.11 km	0.56	0.99
elk-287	aggregate 2	2	0.42	3.77 km	0.56	0.99
elk-363	aggregate 1	1	0.75	0.41 km	0.50	0.89
elk-363	aggregate 1	2	0.25	1.53 km	0.50	0.89
elk-363	aggregate 2	1	0.76	0.13 km	0.50	0.90
elk-363	aggregate 2	2	0.24	4.44 km	0.50	0.90

S4 Reproduction sequence

From the extracted root of the reproducibility package, the versioned R environment is restored with

```
Rscript -e 'renv::restore()'
```

The empirical analysis is then reproduced with

```
Rscript numerics/scripts/run_elk_application.R
```

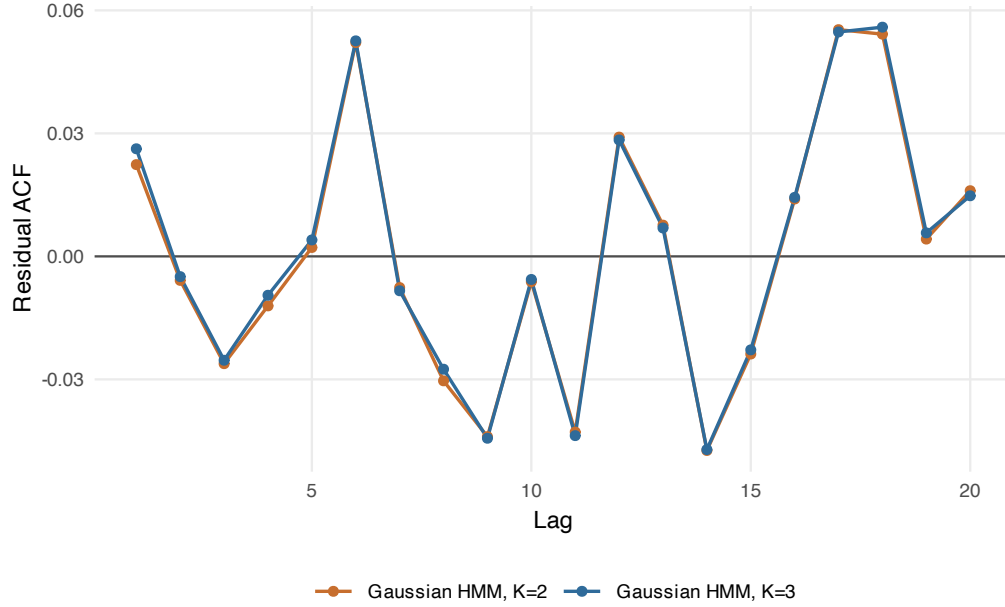


Figure S3: Autocorrelation of approximate one-step standardized prediction residuals for the Gaussian-emission HMMs with $K = 2$ and $K = 3$.

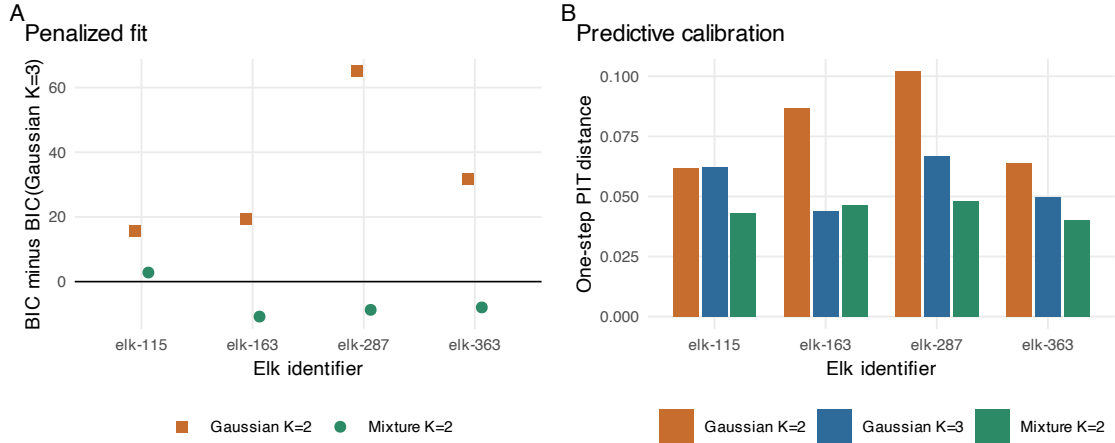


Figure S4: Numerical model comparison for the four elk. Panel A reports BIC differences relative to the three-state Gaussian model; negative values favour the alternative. Panel B reports the Kolmogorov–Smirnov distance between the empirical one-step probability integral transform (PIT) distribution and the uniform distribution; smaller values indicate better descriptive predictive calibration.

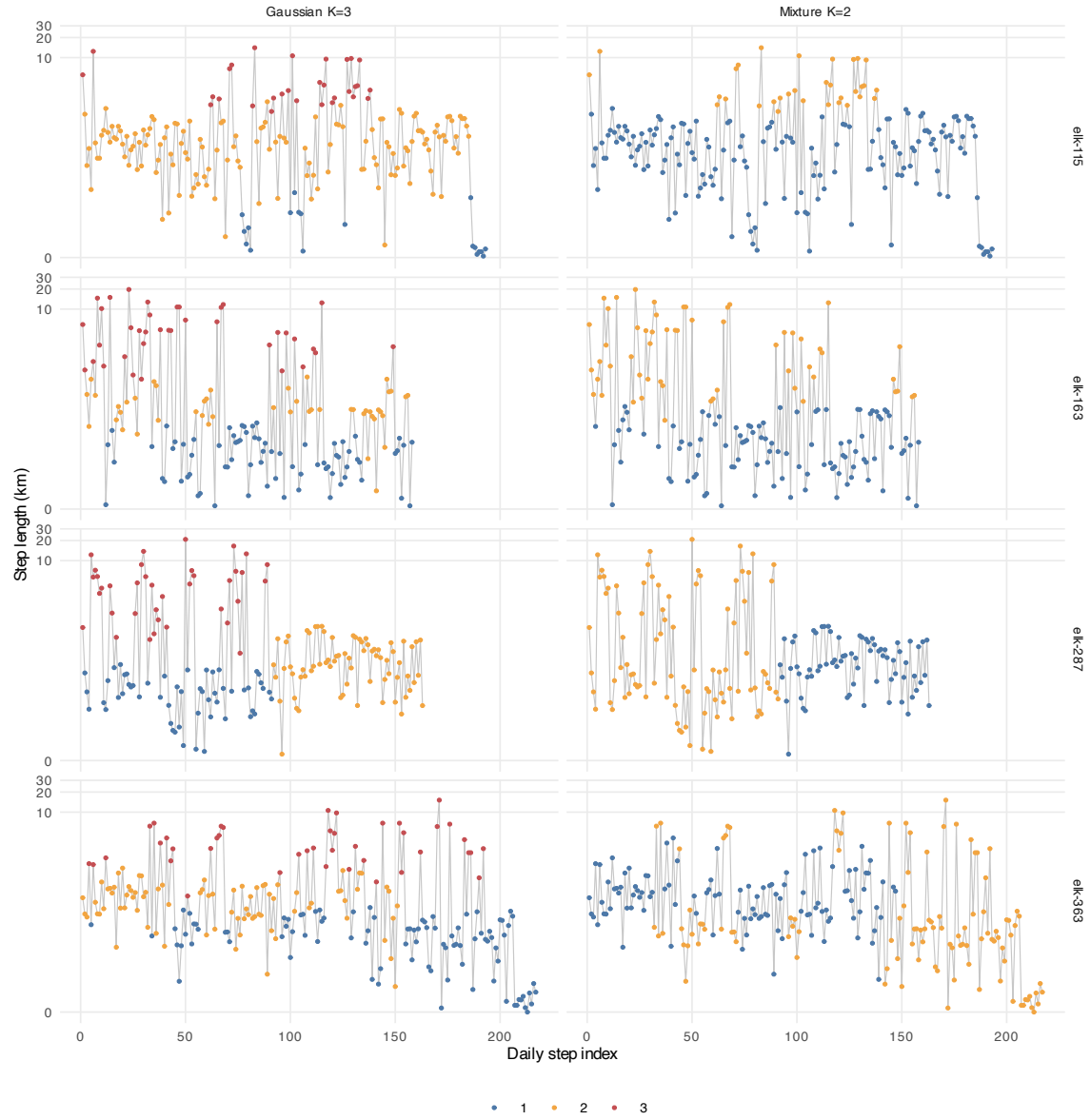
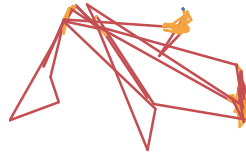


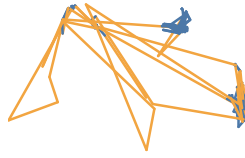
Figure S5: Observed daily step lengths and Viterbi states under the three-state Gaussian and two-state enriched-emission models. Colours denote model-specific statistical states, not assigned behaviours.

Decoded movement paths

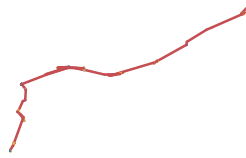
elk-115
Gaussian K=3



elk-115
Mixture K=2



elk-163
Gaussian K=3



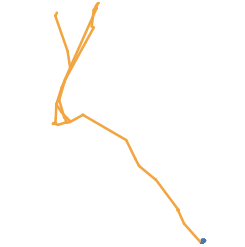
elk-163
Mixture K=2



elk-287
Gaussian K=3



elk-287
Mixture K=2



elk-363
Gaussian K=3



elk-363
Mixture K=2

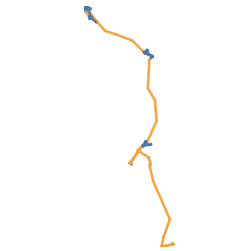


Figure S6: Movement paths coloured by Viterbi state under the three-state Gaussian and two-state enriched-emission models. Long straight segments partly reflect the approximately daily observation schedule and should not be read as fine-scale trajectories.

This command requires R with the `moveHMM` package installed and reads `moveHMM::elk_data` directly. It does not require a separately distributed data file. The fixed-seed simulation study is reproduced with

```
Rscript numerics/scripts/run_review_simulation.R
```

The `renv.lock` file records the package versions used for the archived results. The entire reproducibility workflow, including the custom Gaussian and mixture-emission EM implementations, is written in R. The included `testthat` tests check forward–backward normalization, monotone EM likelihood, transition stochasticity, parameter counts, multi-sequence handling, and the presence and schema of article outputs.